

MEMORY DUMP ANALYZER USING VOLATILITY FRAMEWORK

PROJECT REPORT

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TABLE OF CONTENTS

ABSTRACT.....	3
CHAPTER 1: INTRODUCTION.....	4
1.1 Overview.....	4
1.2 Problem Statement.....	4
1.3 Objectives.....	4
CHAPTER 2: LITERATURE SURVEY.....	5
Volatility Framework.....	5
WinPmem.....	5
MongoDB Atlas.....	5
CHAPTER 3: SYSTEM ANALYSIS.....	6
Existing System.....	6
Limitations.....	6
Proposed System.....	6
Advantages.....	6
CHAPTER 4: SYSTEM DESIGN.....	7
Architecture Diagram.....	7
Modules.....	7
Module 1: Memory Dump Upload.....	7
Module 2: System Information Analysis.....	7
Module 3: Process Analysis.....	8
Module 4: Hidden Process Detection.....	8
Module 5: Risk Scoring Engine.....	8
Module 6: MongoDB Storage.....	8
CHAPTER 5: TECHNOLOGY STACK.....	9
CHAPTER 6: IMPLEMENTATION.....	10
Memory Acquisition.....	10
Volatility Integration.....	10
Process Analysis.....	10
Hidden Process Detection.....	10
Risk Scoring.....	10
MongoDB Integration.....	11
CHAPTER 7: RESULTS.....	12
Sample Results.....	12
CHAPTER 8: FUTURE SCOPE.....	13
CHAPTER 9: CONCLUSION.....	14
REFERENCES.....	15

ABSTRACT

Memory forensics is an essential component of modern cybersecurity investigations. Many malicious activities, including malware execution, hidden processes, credential theft, and unauthorized system access, leave traces only in volatile memory. Traditional disk-based forensic techniques may fail to identify such activities.

The Memory Dump Analyzer is a digital forensic application developed to analyze memory dump files using the Volatility Framework. The system provides an easy-to-use interface that allows users to upload memory dump files and perform forensic investigations. The application extracts system information, analyzes active processes, identifies hidden processes, calculates a risk score based on suspicious activities, and stores results in MongoDB Atlas.

The project integrates React.js for the frontend, Node.js and Express.js for the backend, MongoDB Atlas for storage, WinPmem for memory acquisition, and Volatility 3 for forensic analysis. The developed system demonstrates the practical application of memory forensics in cybersecurity and incident response.

CHAPTER 1: INTRODUCTION

1.1 Overview

Memory forensics is the process of analyzing the contents of a computer's volatile memory (RAM) to identify evidence of malicious activities. Since many modern cyberattacks execute entirely in memory, analyzing memory dumps provides valuable information that may not be available through disk analysis alone.

The Memory Dump Analyzer is designed to automate memory forensic investigations by integrating the Volatility Framework into a web-based application. Users can upload memory dump files and receive detailed forensic analysis results through an intuitive interface.

1.2 Problem Statement

Traditional forensic investigations often focus on disk artifacts. However, modern malware and advanced persistent threats frequently operate in memory and leave minimal traces on disk. Manual memory analysis requires specialized knowledge and is time-consuming.

The challenge is to provide an automated solution capable of:

- Processing memory dump files.
- Extracting process information.
- Identifying suspicious activities.
- Detecting hidden processes.
- Presenting results in an understandable format.

1.3 Objectives

The objectives of the project are:

- To analyze memory dump files using Volatility.
- To extract system information.
- To identify active processes.
- To detect hidden processes.
- To calculate risk scores based on suspicious findings.
- To store analysis results in MongoDB Atlas.
- To provide an interactive dashboard for forensic analysis.

CHAPTER 2: LITERATURE SURVEY

Memory forensics has become increasingly important in cybersecurity investigations. Several tools have been developed for memory analysis.

Volatility Framework

Volatility is one of the most widely used open-source memory forensic frameworks. It supports memory analysis across multiple operating systems and provides numerous plugins for forensic investigations.

WinPmem

WinPmem is a memory acquisition tool used to capture physical memory from Windows systems. It generates memory dump files that can be analyzed using forensic tools.

MongoDB Atlas

MongoDB Atlas provides a cloud-based database platform for storing and managing analysis results.

The proposed system combines these technologies into a unified platform capable of automating memory forensic investigations.

CHAPTER 3: SYSTEM ANALYSIS

Existing System

Existing memory forensic tools often require users to manually execute commands and interpret outputs. This process is complex and requires extensive forensic knowledge.

Limitations

- Command-line complexity.
- Manual analysis.
- Time-consuming investigations.
- Difficult interpretation of results.

Proposed System

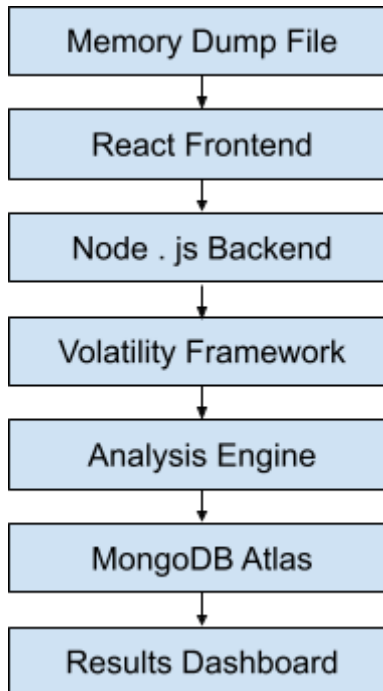
The proposed Memory Dump Analyzer automates forensic investigations by integrating memory acquisition, analysis, risk assessment, and data storage into a single platform.

Advantages

- Automated analysis.
- User-friendly interface.
- Faster investigation process.
- Risk score calculation.
- Cloud database integration.

CHAPTER 4: SYSTEM DESIGN

Architecture Diagram



Modules

Module 1: Memory Dump Upload

Allows users to upload memory dump files for analysis.

Module 2: System Information Analysis

Uses the Volatility plugin:

windows.info - to extract operating system details, kernel information, and memory configuration.

Module 3: Process Analysis

Uses:

windows.pslist - to identify currently active processes.

Module 4: Hidden Process Detection

Uses:

windows.psscan - to scan memory structures and identify hidden processes.

Module 5: Risk Scoring Engine

Detects suspicious processes and calculates a risk score.

Module 6: MongoDB Storage

Stores analysis reports for future reference.

CHAPTER 5: TECHNOLOGY STACK

Component	Technology Used
Frontend	React.js
Backend	Node.js
Framework	Express.js
Database	MongoDB Atlas
Forensic Engine	Volatility 3
Memory Acquisition	WinPmem
API Testing	Browser
Programming Language	JavaScript

CHAPTER 6: IMPLEMENTATION

Memory Acquisition

Memory dumps are generated using WinPmem.

Example:

memory.raw

Volatility Integration

The backend executes Volatility plugins using Node.js.

Examples:

- windows.info
- windows.pslist
- windows.psscan
- windows.cmdline
- windows.netscan

Process Analysis

The application extracts process information from memory and identifies active processes.

Hidden Process Detection

The system compares outputs from pslist and psscan to identify hidden processes.

Risk Scoring

The application maintains a list of suspicious processes including:

- powershell.exe
- cmd.exe
- wmic.exe
- rundll32.exe
- regsvr32.exe

The risk score increases when suspicious processes are detected.

MongoDB Integration

Analysis results are automatically stored in MongoDB Atlas.

CHAPTER 7: RESULTS

The application successfully performs:

- Memory dump upload.
- System information extraction.
- Process analysis.
- Hidden process detection.
- Risk score calculation.
- MongoDB storage.

Sample Results

Active Processes: 198

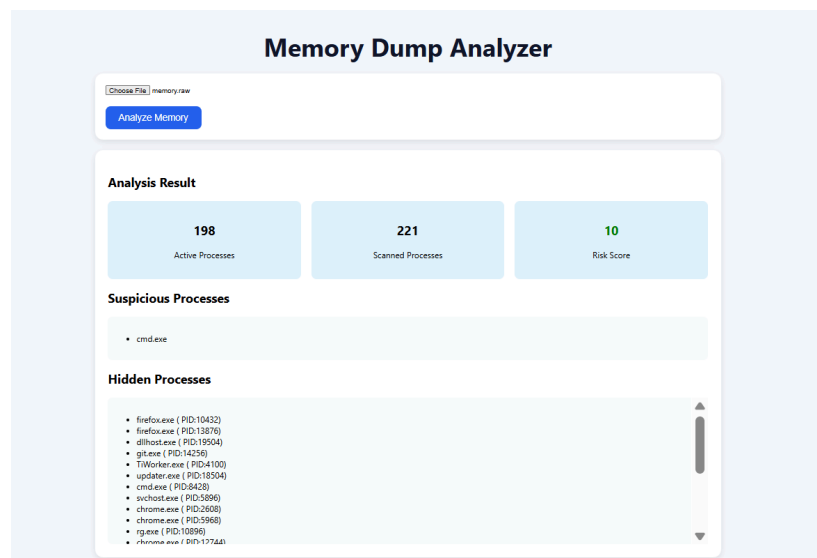
Scanned Processes: 221

Risk Score: 10

Suspicious Process Detected:

cmd.exe

The results demonstrate the effectiveness of memory forensic analysis in identifying suspicious system activities.



CHAPTER 8: FUTURE SCOPE

Future enhancements may include:

- Malware signature detection.
- PDF report generation.
- Real-time memory monitoring.
- Machine learning-based anomaly detection.
- Cloud deployment.
- Threat intelligence integration.

CHAPTER 9: CONCLUSION

The Memory Dump Analyzer successfully automates memory forensic investigations by integrating Volatility, React.js, Node.js, Express.js, and MongoDB Atlas. The application can upload memory dump files, extract system information, identify active and hidden processes, calculate risk scores, and store analysis reports.

The project demonstrates the practical use of memory forensics in cybersecurity investigations and provides a foundation for future enhancements such as malware detection and automated incident response systems.

REFERENCES

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2. WinPmem Documentation.
3. MongoDB Atlas Documentation.
4. Node.js Documentation.
5. Express.js Documentation.
6. React.js Documentation.
7. Digital Forensics and Incident Response Research Papers.